3.9 Social, Cognitive & Neurological Factors in Learning

Learning is the product of the interaction of biological, psychological, and socio-cultural factors.

How does Cognition affect conditioning?

Pavlov & Watson dismissed 'mentalistic concepts' underestimating not only biological factors but also the effects of cognitive processes

Animals can learn an event's predictability :: the reliability of association between CS & UCS

They learn an expectancy :: an anticipation that an event will occur due to presence of specific cues or CS

CCn. treatments ignoring cognition have less success

Ex. using alc spiked w/ nauseating drug as treatment for addiction will lose effectiveness due to person's awareness that nausea comes from the drug

Latent Learning

rats explore a maze with no obvious rewards:
still develop a cognitive map

When reward introduced, these rats run maze as fast as other rats previously reinforced w/ food for this result

ex. of Latent Learning, which becomes apparant only when ∃ incentive to demonstrate it.

Insight Learning

insight learning occurs when we get a sudden flash of insight to solve a problem

little or no systematic interaction w/ environment

What is Observational Learning?

Social learning theory proposes we learn social behavior by observing & imitating others

we do not need to experience consequence ourselves

Observational Learning has limits: observing smne proficient in skill can lead us to over estimate our own abilitis

We learn our native language & other specific behaviors by modeling: process of observing & imitating a specific behavior

Bobo-Doll Experiment

Child watches adult violently attack bobo-doll

Child taken to room w/ many nice toys then is told these are for other kids

taken to room w/ few toys, incl. bobo-doll

child often lashed out at the bobo-doll.

imitated the same acts & used the same words they heard.

also displayed 'non-modeled' aggressive behaviors, ex. assaults w/ dart guns, fights w/ toy animals

this was not seen in children who hadn't seen adult attack bobo doll

Bandura suggested by watching models we experience vicarious conditioning - vicarious reinforcement or vicarious punishment

We learn to anticipate a behaviours consequence in situations like one we are observing

We're esp likely to learn from those we perceive as powerful, successful, or similar to ourselves

fMRI scans show when we see someone get a reward (esp. someone likeable, similar) our own reward systems get activated

Even learned fears can be diminished by watching someone navigate feared situation

Is there a neural basis for Obs. learning?

experiment: Monkeys w/ electrodes in brain area for planning & during movements

these electrodes buzzed when monkey moved

researcher raised ice cream cone with intention to lick electrodes buzzed, and the same thing when monkey watched other monkeys eat

They explained this w/ mirror neurons, which provide neural basis for learning.

Children, even infants, imitate a lot

age 8-16 months: imitate novel gestures
age 12 months: look where adult is looking
age 14 months: imitate acts from TV

age 2-5 year: children overimitate, copying irrelevant adult actions

Our observational learning makes emotions contagious

because of how strong obs. learning is, we may misremember action observed as one we have performed (source amnesia: source misattribution)

Observing others' pain releases our own painkillers

What is the impact of modeling?

Prosocial impact

modeling prosocial behavior can have prosocial effects

parents are powerful models

Obs. learning of morality begins early

models are most effective when actions + words are consistent

exposed to hypocrite → kids imitate hypocrisy

Antisocial Impact

Obs. learning can also increase antisocial behavior

MCQ

1. c 2. c
3. d 4. c
5. d 6. c

Personal Connection

~2 weeks ago, on a calc test I was having trouble with a question, so I came back to it after and after I had a sudden insight